

LAYER 01 Author

Source Modules

Dataset components
Task/analysis components



Compiler

Validates & packages
each source module



Compiled Modules

Reusable structured units



LAYER 02 Build

Lesson Manifests

Defines which modules
combine into each lesson



Build Script

Assembles modules into
complete lesson scripts



jupytertext

Student Notebooks

Jupyter .ipynb files



LAYER 03 Run

IPython Kernel

Runs student code
Hooks into each cell event



Telemetry Collector

Captures events:
revisions, errors, idle time



Process Logs

Stored locally per session



Teacher Dashboard



Real-time alerts &
interventions

→ Synchronous
...→ Async / optional